FREIGE: Faithful Evidence Generation for Document-Level Relation Extraction via Rejection Sampling Fine-Tuning

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Abstract

Large language models (LLMs) have approached encoder-based methods in document-level relation extraction (DocRE) accuracy, yet no existing LLM method generates verifiable evidence—the supporting sentences that justify each extracted relation. We present FREIGE, the first LLM-based DocRE system that jointly produces relation triples and faithful sentence-level evidence. Our approach uses **Rejection Sampling Fine-Tuning (RSFT)**: an SFT-trained model generates candidate outputs, a Contrastive Evidence Discrimination (CED) scorer filters high-quality candidates using document-internal hard negatives, and the filtered data is used for further fine-tuning. On DocRED, FREIGE achieves test rel_F1 of 43.2% (+3.2pp over SFT, $p<0.01$) and joint Evi-F1 of 32.2%, while producing human-auditable sentence citations. We further demonstrate that online reward optimization (GRPO) with evidence-quality rewards causes catastrophic *evidence-extraction decoupling*: evidence quality rises while extraction performance collapses to near-zero, and this failure *accelerates* with larger group sizes that provide more precise advantage estimation. We introduce EDCR (Evidence Distractor Citation Rate) as a complementary diagnostic to Evi-F1 and validate it through human annotation ($r=0.44$, perfect method-level rank concordance). Our results establish that for structured IE tasks, evidence-quality signals are most effective as offline data curation criteria rather than online training rewards.

1 Introduction

Document-level relation extraction (DocRE) requires identifying relations between entities across multiple sentences in a document (Yao et al., 2019). While encoder-based methods have achieved strong performance (Zhou et al., 2021; Ma et al., 2023; Xiao et al., 2022), LLM-based approaches have recently emerged as competitive alternatives (Xu et al., 2024), offering the advantage of flexible generation. However, no existing LLM-based DocRE method produces *verifiable evidence*—the specific sentences supporting each extracted relation—despite this being critical for deployment in human-in-the-loop knowledge base population, biomedical curation, and RAG provenance tracking.

Encoder-based methods like DREEAM (Ma et al., 2023) derive evidence from attention weights, producing implicit, post-hoc explanations. In contrast, a generative approach can directly output explicit sentence indices alongside each triple, enabling human-auditable evidence that is fundamentally more interpretable. The challenge is teaching the model to cite *faithfully*: naive supervised fine-tuning (SFT) produces models where 71.2% of entity-sharing citations are distractor sentences that mention relevant entities but do not actually support the predicted relation.

A natural approach to improve evidence quality is reinforcement learning with NLI-based faithfulness rewards. We explore this direction and discover a fundamental obstacle: **evidence-extraction decoupling**. Under GRPO training with evidence-quality rewards, the model rapidly learns to produce higher-quality evidence (evi_F1 rising to 0.835–0.862) while simultaneously losing the ability to extract relations (rel_F1 collapsing to <0.08). This occurs across all reward types (CED, flat-NLI, extraction-only) and learning rates, and *worsens* with larger group sizes (G=8 vs. G=4) that provide more precise advantage estimation—directly connecting to reward overoptimization theory (Gao et al., 2023).

Our solution is **FREIGE** (**F**aithful **R**elation **E**xtraction with **I**nference-time **G**eneration and **E**vidence), built on Rejection Sampling Fine-Tuning (RSFT). Instead of using evidence-quality signals as online rewards, we use them as offline

data curation criteria: an SFT model generates N candidate outputs per document, a Contrastive Evidence Discrimination (CED) scorer ranks candidates by evidence quality using document-internal hard negatives, and only the top-ranked candidate is retained for further fine-tuning. This circumvents the decoupling failure entirely—the model never receives reward gradients, only learns from curated high-quality demonstrations.

Our contributions are:

1. We present the first LLM-based DocRE method that jointly generates relation triples and verifiable sentence-level evidence, achieving +3.2pp rel_F1 over SFT on DocRED test ($p<0.01$) with consistent gains across three model sizes (1.7B/4B/8B) and on Re-DocRED.

2. We characterize evidence-extraction decoupling in GRPO+QLoRA for document-level IE, showing it accelerates with improved advantage estimation (G=8 peaks earlier and collapses faster than G=4), and that offline preference learning (DPO) partially mitigates but does not solve the underlying issue.

3. We introduce EDCR as a diagnostic metric complementary to Evi-F1, validated through human annotation (r=0.44, method-level rank concordance 1.0).

2 Related Work

Document-Level Relation Extraction. DocRE extends sentence-level RE to require cross-sentence reasoning (Yao et al., 2019). Encoder-based methods dominate, with ATLOP (Zhou et al., 2021), DREEAM (Ma et al., 2023), and SAIS (Xiao et al., 2022) achieving strong performance through adaptive thresholding and evidence-aware attention. Recent LLM-based methods include LMRC (Xu et al., 2024), which achieves 60.7% F1 with LLaMA3.1-8B but produces no evidence. Our work is the first to address evidence generation in generative DocRE.

Evidence and Faithfulness in IE. DREEAM (Ma et al., 2023) uses attention supervision for evidence extraction, achieving 57.3% joint Evi-F1. SAIS (Xiao et al., 2022) provides sentence importance scores (55.7% joint Evi-F1). Both rely on attention-derived evidence, which is implicit and not directly auditable. Our approach generates explicit sentence citations, enabling transparent verification.

Rejection Sampling and Offline RL. Rejection sampling fine-tuning (RSFT/ReST) has been shown theoretically equivalent to PPO under certain conditions (Dong et al., 2024). We apply RSFT to structured IE, using CED as the filtering criterion rather than task reward. Our findings connect to reward overoptimization (Gao et al., 2023) and the evidence grounding gradient vanishing analysis of LongRLVR (Song et al., 2025).

RL for Information Extraction. Prior work has applied RL to IE tasks with mixed results. Our systematic comparison across the data curation (RSFT) vs. preference learning (DPO) vs. reward optimization (GRPO) spectrum provides empirical evidence that structured generation tasks are particularly vulnerable to reward-based training.

3 Method

3.1 Task Formulation

Given a document $D = \{s_1, \ldots, s_n\}$ with entity mentions, the model generates a set of tuples $\{(h_i, r_i, t_i, E_i)\}$ where h_i, t_i are head/tail entities, r_i is a relation type, and $E_i \subseteq \{s_1, \ldots, s_n\}$ is the supporting evidence (sentence indices).

3.2 SFT Baseline

We fine-tune Qwen3-4B with QLoRA on DocRED training data formatted as structured generation: the input is the document with entity markers, and the output is a linearized list of (head, relation, tail, [evidence sentence indices]) tuples.

3.3 Contrastive Evidence Discrimination (CED)

For each predicted triple (h, r, t, E), CED scores evidence quality by comparing the entailment probability of cited sentences against entity-sharing distractor sentences:

$$\text{CED}(h, r, t, E) = \frac{1}{|E|} \sum_{s_i \in E} \left[p_{\text{NLI}}(s_i \Rightarrow r) - \max_{s_j \in \mathcal{N}} p_{\text{NLI}}(s_j \Rightarrow r) \right] \quad (1)$$

where $\mathcal{N}$ is the set of entity-sharing non-cited sentences (document-internal hard negatives) and p_{NLI} is the entailment probability from a DeBERTa-v3-base NLI model.

The contrastive margin penalizes citations that are not more entailing than the best available distractor, addressing the entity-mention bias where sentences sharing entities with the target triple receive inflated NLI scores regardless of actual support.

3.4 Rejection Sampling Fine-Tuning (RSFT)

The RSFT pipeline operates in three stages:

1. **Candidate Generation**: The SFT model generates $N{=}8$ candidate outputs per document via temperature sampling.

2. **CED Scoring and Filtering**: Each candidate is scored by CED. Only the top-1 candidate per document is retained (selection rate $\sim$12.4%, yielding 2,718 training samples).

3. **Fine-tuning**: The filtered high-quality candidates serve as training data for additional SFT epochs.

This design avoids modifying the policy gradient—the model learns only from curated demonstrations, not from reward signals. The key insight is that CED's value lies in *ranking* candidates rather than providing absolute quality scores, making the top-k selection robust to scorer calibration errors.

3.5 EDCR: Evidence Distractor Citation Rate

We propose EDCR as a complementary diagnostic to Evi-F1:

$$\text{EDCR} = \frac{|\{s \in \text{Cited} : s \in \mathcal{N}_{\text{entity-sharing}} \setminus \text{Gold}\}|}{|\text{Cited}|} \quad (2)$$

EDCR measures the proportion of cited sentences that share entities with the target triple but are not in the gold evidence set. While Evi-F1 measures evidence recall (whether gold evidence is cited), EDCR measures evidence precision in terms of distractor contamination.

4 Experimental Setup

4.1 Data and Evaluation

We use DocRED (Yao et al., 2019) as the primary benchmark and Re-DocRED (Tan et al., 2022) for generalization evaluation. We report: (1) **rel_F1**: relation extraction F1; (2) **Evi-F1$_{\text{joint}}$**: joint evidence F1 following the DREEAM protocol (Ma et al., 2023), computed over all gold triples; (3) **Evi-F1$_{\text{cond}}$**: conditional evidence F1 over correctly predicted triples; (4) **EDCR**: evidence distractor citation rate.

Method	rel_F1	ign_F1	Evi-F1$_{\text{j}}$	Evi-F1$_{\text{c}}$
DREEAM†	62.8	—	57.34	∼93
SAIS†	60.4	—	55.67	∼92
LMRC†	60.7	—	—	—
SFT (Qwen3-4B)	40.0	39.8	29.5	79.0
RSFT+flat-NLI	42.5	42.3	31.2	79.5
RSFT+CED (FREIGE)	**43.2**	**43.0**	**32.2**	**80.0**

Table 1: Test set results on DocRED. †Published results. Evi-F1$_{\text{j}}$: joint protocol (DREEAM-aligned); Evi-F1$_{\text{c}}$: conditional on correct triples. FREIGE achieves +3.2pp rel_F1 over SFT ($p{<}0.01$).

4.2 Models and Training

Our primary model is Qwen3-4B with QLoRA ($\alpha{=}128$, rank 64). For model size ablation, we additionally evaluate Qwen3-1.7B and Qwen3-8B. CED scoring uses DeBERTa-v3-base fine-tuned on MNLI. RSFT uses $N{=}8$ candidates per document, top-$k{=}1$ selection.

4.3 Baselines

We compare against: (1) **SFT**: our Qwen3-4B SFT baseline; (2) **DREEAM** (Ma et al., 2023): RoBERTa-large encoder (reproduced, test Evi-F1$_{\text{joint}}$=57.34%); (3) **SAIS** (Xiao et al., 2022): RoBERTa-large encoder (reproduced, test Evi-F1$_{\text{joint}}$=55.67%); (4) **LMRC** (Xu et al., 2024): LLaMA3.1-8B generative method (F1=60.7%, no evidence). We also evaluate intermediate methods: **GRPO+CED** (online reward optimization), **DPO+CED** (offline preference learning), **CED-as-reranker** (inference-time reranking), and **RSFT+flat-NLI** (non-contrastive scoring ablation).

5 Results

5.1 Main Results

Table 1 presents test set results on DocRED.

FREIGE achieves 43.2% rel_F1, a +3.2pp improvement over SFT ($p{<}0.01$), and is the first LLM method to report joint Evi-F1 (32.2%). The gap to encoder methods in joint Evi-F1 (32.2 vs. 57.3) is primarily driven by extraction recall (43.2% vs. 62.8%) rather than evidence quality: conditional Evi-F1 (80.0% vs. ∼93%) shows a much smaller gap, and our explicit sentence

Config	Scorer	rel_F1	evi_F1	EDCR
r1 (3ep, α128, s42)	CED	44.49	81.87	65.65
	flat-NLI	43.98	81.15	67.25
s43 (3ep, α128, s43)	CED	46.83	81.56	63.88
	flat-NLI	46.10	80.78	65.90
r2b (1ep, α32, lr1e-5)	CED	44.93	81.28	66.09
	flat-NLI	44.70	80.53	67.52
SFT baseline	—	40.57	79.79	71.18

Table 2: CED vs. flat-NLI RSFT ablation (dev set). CED outperforms flat-NLI in the same direction across all three configurations, but differences are not statistically significant.

Config	Best rel_F1	Peak	evi_F1$_{\text{end}}$	Pattern
G=4, kl.01	21.7	s100↑	57.5	Slow climb
G=8, kl.01	7.8	s40	86.2	Peak→collapse

Table 3: GRPO G=4 vs. G=8. Larger group size provides more precise advantage estimation but accelerates decoupling.

Step	rel_F1	evi_F1	EDCR	Reward
10	0.8	72.0	98.5	0.62
20	3.5	76.0	97.2	0.85
30	6.2	79.5	95.0	1.05
40	**7.8**	83.5	92.8	1.18
50	5.5	84.5	93.5	1.28
60	3.8	85.0	94.0	1.35
80	2.0	85.8	94.5	1.42
100	1.2	86.2	94.8	1.45

Table 4: GRPO G=8 checkpoint trajectory. Reward increases monotonically while rel_F1 peaks then collapses, exhibiting evidence-extraction decoupling.

Method	Type	rel_F1	evi_F1	EDCR	ΔSFT
GRPO G=8	Online reward	7.8	83.5	92.8	−32.8
GRPO G=4	Online reward	21.7	57.5	85.4	−18.9
DPO+CED	Offline pref.	34.0	79.8	66.0	−6.6
SFT	Supervised	40.6	79.8	71.2	—
RSFT+flat-NLI	Data curation	46.1	80.8	65.9	+5.5
RSFT+CED	Data curation	46.8	81.6	63.9	+6.3

Table 5: Training paradigm comparison (dev set). Data curation (RSFT) succeeds where both online (GRPO) and offline (DPO) contrastive losses fail. ΔSFT: rel_F1 change in pp.

citations are directly auditable unlike attention-derived evidence.

5.2 CED vs. Flat-NLI Scoring (Ablation)

Table 2 compares CED and flat-NLI scoring for RSFT on the dev set.

CED outperforms flat-NLI consistently across all three hyperparameter configurations in both evi_F1 (+0.75pp mean) and rel_F1 (+0.49pp mean), and achieves lower EDCR (−1.68pp mean). However, paired bootstrap tests (10,000 resamples) show these differences are not statistically significant (evi_F1: p=0.12; rel_F1: p=0.28; EDCR: p=0.09).

Critically, *both* CED and flat-NLI RSFT substantially outperform SFT: flat-NLI RSFT achieves +5.5pp rel_F1 over SFT (p<0.001), establishing that the RSFT pipeline itself—regardless of scoring function—is the primary contribution. The 78% overlap in selected candidates between the two scorers explains the limited differentiation: DeBERTa-v3-base's positional bias on long documents causes contrastive and non-contrastive scores to converge in high-confidence regions.

5.3 Evidence-Extraction Decoupling in GRPO

Table 3 summarizes the GRPO results. With G=8, rel_F1 peaks at step 40 (7.8%) then collapses to 1.2% by step 100, while evidence quality monotonically rises to 86.2%. This is *worse* than G=4 (best rel_F1=21.7%), decisively ruling out the hypothesis that G=4 was simply too small for stable training.

Table 4 shows the full G=8 trajectory, revealing the characteristic signature of evidence-extraction decoupling: reward increases monotonically ($0.62 \rightarrow 1.45$) while rel_F1 rises briefly then collapses.

The mechanism is: more precise advantage estimation (lower variance $\propto 1/G$) enables the model to more efficiently optimize the CED reward, which measures evidence quality but is misaligned with relation extraction. The "success" of G=4 (rel_F1=21.7) is an accidental byproduct of noisy advantage estimation acting as implicit regularization, analogous to the role of dropout in preventing overfitting.

Across all GRPO configurations, reward and rel_F1 show strong negative correlation (r=−0.68), while reward and evi_F1 are strongly positively correlated (r=0.82). The decoupling onset occurs at step ∼30 for G=8 versus step ∼60 for G=4, consistent with faster optimization under lower-variance advantage estimates.

Method	rel_F1	evi_F1$_c$	EDCR	Zone
GRPO G=8	7.8	83.5	92.8	Collapse
GRPO G=4	21.7	57.5	85.4	Collapse
SFT	40.6	79.8	71.2	Baseline
CED-reranker	42.2	80.9	66.8	Filtered
DPO+CED	34.0	79.8	66.0	Filtered
RSFT+flat-NLI	46.1	80.8	65.9	Filtered
RSFT+CED	46.8	81.6	63.9	Filtered

Table 6: Cross-method EDCR diagnostic (dev set). EDCR discriminates three zones: collapse (>85%), baseline (∼71%), and filtered (64–67%).

Statistic	Value
Samples / Annotators	100 / 2
Inter-annotator κ	0.62
EDCR vs. human Pearson r	0.44
EDCR vs. human Spearman ρ	0.41
EDCR overestimate	+8.5pp
False distractor rate	14%
Method-level rank concordance	1.0

Table 7: EDCR human validation statistics.

top-k	N_{train}	rel_F1	evi_F1	EDCR
1	2,718	**46.8**	**81.6**	**63.9**
3	8,154	45.8	80.2	66.2

Table 8: Top-k selection ablation (dev set). Strict filtering (top-1) outperforms relaxed filtering (top-3) despite 3× less training data.

5.4 Data Curation vs. Preference/Reward Learning

Table 5 reveals a clear ordering across training paradigms. DPO (β=0.1, 5,436 preference pairs) avoids GRPO's catastrophic collapse but still degrades rel_F1 by 6.6pp below SFT. The DPO contrastive loss simultaneously pushes up preferred and pushes down rejected candidates; in structured IE, "rejected" candidates may contain correct relation patterns with merely weaker evidence, so pushing them down damages extraction capability. DPO is also more sensitive to CED label noise (∼18.5% error rate) than RSFT, which only uses top positive candidates and never explicitly penalizes negatives.

The key insight is that the distinction is not online vs. offline RL, but *contrastive/reward-based loss vs. data curation*: any method that modifies gradients based on evidence-quality comparisons risks damaging structured generation, while RSFT succeeds precisely because it is not RL—it is quality-filtered SFT.

5.5 EDCR as Diagnostic Metric

Table 6 demonstrates EDCR's ability to discriminate three method categories. EDCR correlates negatively with rel_F1 (ρ=−0.52) and evi_F1 (ρ=−0.38) across methods, capturing the intuition that better methods produce fewer distractor citations.

Human Validation. We validate EDCR through a 100-sample annotation study with two annotators (κ=0.62). Table 7 summarizes the results. EDCR achieves Pearson r=0.44 and Spearman ρ=0.41 against human judgments at the sample level. While modest, method-level rank concordance is perfect (1.0), confirming EDCR reliably discriminates between methods. EDCR systematically overestimates the distractor rate by 8.5pp, primarily because 14% of sentences labeled as distractors by the automatic metric are judged by annotators as actually supporting the relation—reflecting incomplete gold evidence annotations in DocRED rather than metric failure.

5.6 CED as Inference-Time Reranker

Without any additional training, CED can be used to rerank N=8 SFT candidates at inference time. CED-reranker achieves rel_F1=42.2% (+1.6pp over SFT) and EDCR=66.8% (−4.4pp over SFT) on 985 documents, but at 8.5× inference cost. Compared to RSFT's +6.3pp rel_F1 improvement at no additional inference cost, reranking demonstrates that CED signals are more effectively transmitted through training data curation than inference-time selection.

5.7 Top-k Selection Ablation

Table 8 shows that top-k=1 (strict filtering, 2,718 samples) uniformly outperforms top-k=3 (relaxed filtering, 8,154 samples). The mean CED score drops from 0.668 to 0.512 when including rank-2/3 candidates, and the lower-quality candidates introduce distractor citation patterns that degrade evidence quality (EDCR +2.3pp). This validates the RSFT design principle: quality of curated data matters more than quantity.

5.8 Model Size Ablation

Table 9 demonstrates consistent RSFT improvement across three model sizes (+2.8/+6.3/+4.0pp). Smaller models generate fewer high-quality candidates, reducing RSFT training data and limiting gains. The largest absolute performance is

Model	Method	rel_F1	evi_F1$_c$	Δrel_F1
Qwen3-1.7B	SFT	32.0	77.5	—
	RSFT+CED	34.8	79.2	+2.8pp
Qwen3-4B	SFT	40.6	79.8	—
	RSFT+CED	46.8	81.6	+6.3pp
Qwen3-8B	SFT	44.8	81.0	—
	RSFT+CED	48.8	82.5	+4.0pp

Table 9: Model size ablation (dev set). RSFT consistently improves across 1.7B, 4B, and 8B.

Method	DocRED	Re-DocRED	ΔSFT	EDCR
SFT	40.6	48.5	—	66.8
RSFT+CED	46.8	52.0	+3.5pp	61.0

Table 10: Re-DocRED evaluation (dev set, DocRED-trained models). RSFT improvement transfers (+3.5pp).

Method	rel_F1	Evi-F1$_j$	Evi-F1$_c$
DREEAM (repro.)	62.5	58.1	∼93
SAIS (repro.)	60.8	56.2	∼92
RSFT+CED (ours)	46.8	37.2	81.6
SFT (ours)	40.6	30.8	79.8

Table 11: Dev set comparison with reproduced baselines (DREEAM-aligned joint Evi-F1 protocol). Reproduction gap $<$1pp from published test results.

NLI Scorer	Disc. Acc.	rel_F1	evi_F1	EDCR
DeBERTa-v3-base	81.5	46.8	81.6	63.9
DeBERTa-v3-large	84.8	47.2	82.0	63.2

Table 12: NLI model ablation. Upgrading the scorer (+3.3pp discrimination accuracy) yields minimal downstream gains (+0.4pp rel_F1), indicating the pipeline bottleneck is candidate generation quality, not scorer precision.

Qwen3-8B RSFT (rel_F1=48.8%), though a gap to LMRC (60.7%, 8B, no evidence) remains, reflecting the additional burden of evidence generation.

5.9 Re-DocRED Generalization

Models trained on DocRED and evaluated on Re-DocRED (which corrects ∼30% false negative triples) show consistent RSFT improvement (+3.5pp, Table 10). The delta narrows from +6.3pp on DocRED to +3.5pp on Re-DocRED, a ceiling effect: Re-DocRED's more complete annotations strengthen the SFT baseline, compressing the improvement margin.

5.10 Baseline Reproduction

We reproduce DREEAM and SAIS on the DocRED dev set to validate evaluation protocol alignment (Table 11).

Reproduced dev results are within 1pp of published test results (DREEAM: 58.1 vs. 57.3; SAIS: 56.2 vs. 55.7), confirming evaluation alignment. The joint Evi-F1 gap (37.2 vs. 58.1) decomposes as: joint $\approx$ recall $\times$ conditional $\approx 0.47 \times 0.82 = 0.385$, showing the gap is driven by extraction recall rather than per-triple evidence quality.

5.11 NLI Model Ablation

Upgrading the NLI scorer from DeBERTa-v3-base to DeBERTa-v3-large improves CED discrimination accuracy by 3.3pp (81.5%→84.8%) but yields only +0.4pp downstream rel_F1 (Table 12). The pipeline bottleneck is candidate generation quality (fixed at N=8 SFT samples), not scorer precision; base-level accuracy (81.5%) already exceeds the threshold for effective filtering.

6 Discussion

Why RSFT Succeeds Where RL Fails. The consistent failure of reward-based methods (GRPO, DPO) and success of data curation (RSFT) reveals a structural property of evidence-generating IE: the evidence-quality objective is misaligned with the extraction objective at the gradient level. Reward-based methods transmit this misalignment directly through policy gradients, while RSFT insulates the model by converting the quality signal into a data selection criterion. This connects to the theoretical result that RSFT is equivalent to PPO under certain conditions (Dong et al., 2024)—but with a crucial practical advantage: RSFT eliminates the risk of reward overoptimization because the model never optimizes a reward function.

The Role of CED. Our ablation shows CED's contrastive margin provides consistent but statistically non-significant improvements over flat-NLI scoring (p=0.12 for evi_F1). This is attributable to three factors: (1) DeBERTa-v3-base's positional bias causes contrastive and non-contrastive scores to converge for long documents; (2) incomplete DocRED gold evidence (∼14% false distractors) dilutes the hard-negative signal; (3) the indirect RSFT pipeline (scorer → filter → training data → fine-tune) attenuates scoring-level differences. Importantly, CED's top-1 vs. top-3 selection ad-

vantage (Table 8) demonstrates that the *ranking* quality of the scorer, even if not differentiated from flat-NLI, enables effective data curation.

EDCR Limitations. While EDCR achieves perfect method-level rank concordance with human judgments, its sample-level correlation (r=0.44) is moderate. The 14% false distractor rate—where automatically labeled distractors are judged by humans as valid evidence—reflects DocRED's annotation incompleteness rather than metric design failure. EDCR should be used as a complementary diagnostic alongside Evi-F1, not as a standalone metric.

7 Limitations

Absolute performance gap. Our best model (Qwen3-8B RSFT, rel_F1=48.8%) trails encoder methods (DREEAM 62.8%) and LLM methods without evidence (LMRC 60.7%). Evidence generation imposes an additional burden on the model; closing this gap while maintaining evidence quality is future work.

Single NLI backbone. CED relies on DeBERTa NLI models, which have known biases including lexical overlap sensitivity and positional bias on long documents. Alternative evidence scorers (e.g., cross-encoder models, LLM-based evaluation) may better differentiate contrastive from non-contrastive scoring.

GRPO scope. Our decoupling findings are established for GRPO+QLoRA on document-level IE. We do not claim this generalizes to full-rank GRPO, other online RL algorithms (PPO, REINFORCE), or other structured generation tasks.

Gold annotation quality. DocRED gold evidence annotations are incomplete (∼14% false distractors by our human study), which affects both training (RSFT candidates scored against noisy gold) and evaluation (EDCR systematically overestimates by 8.5pp).

Dev-to-test degradation. RSFT shows 3.6pp dev-to-test degradation in rel_F1 (vs. 0.6pp for SFT), suggesting indirect overfitting through the CED scoring pathway. The improvement remains significant on test (p<0.01) but this asymmetric degradation warrants investigation.

8 Conclusion

We introduced FREIGE, the first LLM-based method for document-level relation extraction that jointly generates relation triples and verifiable sentence-level evidence. Through systematic comparison of training paradigms, we established that evidence-quality signals are effective as offline data curation criteria (RSFT, +3.2pp test rel_F1) but destructive as online rewards (GRPO) or offline contrastive losses (DPO). We characterized evidence-extraction decoupling in GRPO+QLoRA, showing it accelerates with larger group sizes, and introduced EDCR as a human-validated diagnostic for evidence quality. Our work opens the path toward trustworthy, evidence-grounded document-level IE with large language models.

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